Making Scientific Knowledge Navigable for Agents

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Scientific knowledge is fragmented and weakly linked

We query it as if it were flat text. We can do better.

Knowledge is scattered

Literature, datasets, software, posters, talks — in different systems, with different identifiers, and no common seams.

The structure we have is incomplete

DOIs, citation graphs, data-availability statements, ORCIDs — partial coverage, inconsistent metadata, ambiguous attribution.

Relationships are underspecified

A citation can mean: builds on, disproves, mentions, reuses data from, or reuses methods from. The graph collapses all five into one edge.

Search is the universal API

Keyword or vector search returns a ranked list. That's fine for 'find me a paper.' It's not enough for 'map the landscape' or 'what breaks if this result is wrong.'

The scale problem: research output accelerates like codebases do

Engineers gave their agents structure. Scientists haven't — yet.

3M+

scientific papers published per year

doubling every ~12 years

20–30%

of a researcher's time

spent on literature search

AI-accelerated

both code and research output

the same growth curve hits both

Codebases outran human reading years ago. Engineers responded by giving agents structure — dependency graphs, call graphs, blast-radius tools. Scientific literature has the same shape of problem and has not yet had the same response.

Fragmentation of knowledge: serendipity over systems

Right now, transferable methods travel by chance. That's the thing to fix.

Citations encode more than dependency

Prestige, convention, self-citation, availability bias. A clean technical DAG is not what the citation graph actually is.

Connections happen through serendipity, not systems

Whether two lines of work meet depends on who read what, when. It is not a structural property of the knowledge itself.

Disciplinary silos hide transferable methods

A technique useful in one field may have been solved in a neighbouring one years earlier — under different vocabulary.

The shift worth making

Turn serendipitous discovery into retrievable relevance — so an agent can surface it, and a researcher can rely on it.

Codebases as analogy

Software stopped treating code as isolated files. Science can do the same.

Software engineering	Science
Dependencies what code relies on to run	**Reused methods and datasets** what a result relies on to hold
Call graphs chains of function calls	**Chains of reasoning** a claim that rests on a claim that rests on a dataset
Blast radius what breaks if this changes	**Result impact** which follow-on findings are at risk if this one is wrong

Citation graphs approximate all of this — but they're incomplete, semantically weak, and socially biased. The real signal is the underlying ecosystem of data, methods, results, code, and claims.

Toward 'debugging' science

Treat outputs as claims to evaluate — independent of citation count or venue prestige.

Science already has implicit acceptance criteria

- Physical consistency — does it fit known laws?
- Reproducibility — can someone else hit the same result?
- Compatibility — does it agree with adjacent findings?

What a navigable layer unlocks

Automated consistency checks

Flag claims that contradict well-established adjacent results.

Conflicting-result detection

Surface studies that disagree rather than burying the tension in citations.

Missing-link identification

Point out papers that should cite each other but don't — and why.

Method-reuse surfacing

Find transferable techniques across silos, not just nearest-neighbor prose.

What "navigable" means for an agent

Agents benefit from ranked lists too — they just benefit from more.

Human searcher

Ranked list

Scans titles, opens a few, satisfied — same primitive an agent uses for cheap queries.

One-shot query

Session state lives in the researcher's head.

Visual cues

Journal, authors, year, formatting.

Stops when tired

Implicit budget, intuition-driven.

Agent (given the right primitives)

Graph topology

Which papers bridge communities? Which are structurally central?

Session working set

Accumulates, dedupes, tags, reasons across turns.

Structured returns

Bibcodes, IDs, community labels, provenance — composes with other tools.

Multi-hop moves

Citation chains, co-citation, bridge-paper gaps, temporal.

Vector search and knowledge graphs, together

Hybrid retrieval is the table stakes. Agent-mediated iteration is the unlock.

Semantic

Dense vectors — domain-tuned plus general-purpose, fused via reciprocal rank.

Lexical

BM25 on titles, abstracts, keywords. Catches specific terms that embeddings blur.

Structural

Graph moves: citation chains, co-citation, shortest paths, community lenses.

Symbolic

Entity and claim extraction — instruments, datasets, methods, results as first-class objects.

Iterative

An agent chooses the next move based on what it just learned. The loop — not any single tool — is where the leverage lives.

Why MCP is a good fit for this layer

Model Context Protocol gives the agent tools it can actually hold.

Tool-calling primitives

First-class: name, typed args, typed returns. The agent knows what it can do.

Structured responses

JSON objects, not HTML snippets. Compose cleanly with other tools.

Portable discipline

If your research copilot speaks MCP, the same knowledge layer can be hosted by ADS, arXiv, your own group, or someone else.

Session context

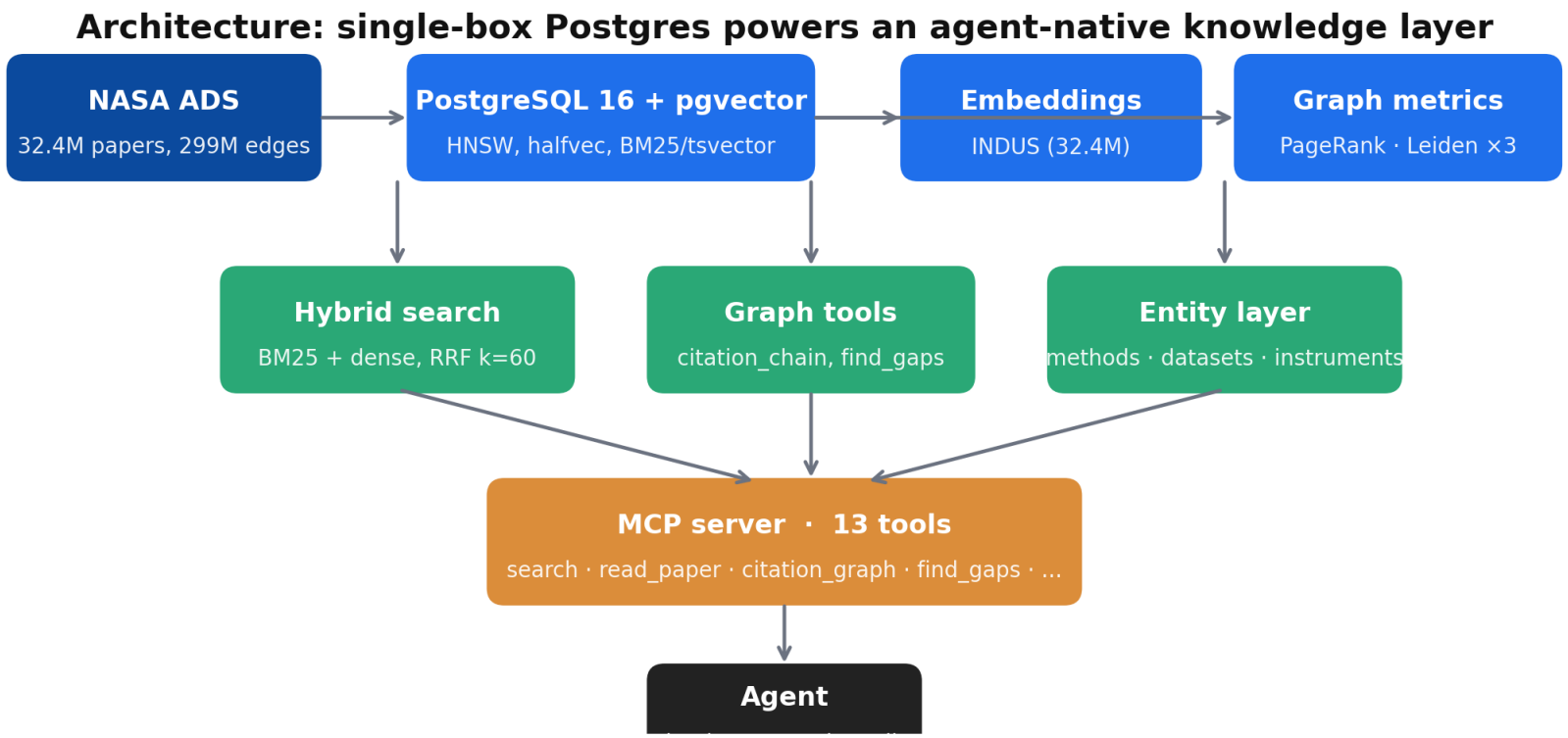
Stateful per-connection. Working sets live on the server, not in the prompt.

Transport-agnostic

stdio, HTTP, WebSocket. Works with Claude Desktop, Cursor, custom agents.

A working prototype: SciX on NASA ADS

32.4M papers, PostgreSQL + pgvector, single server, one MCP endpoint.



Deliberate simplicity: no separate vector DB, no orchestrator, no queue. Postgres + pgvector carries the corpus, the graph metrics, and the retrieval stack.

Two vector stores, two jobs

Pick the backend for the move — the agent only sees one MCP interface.

pgvector — the corpus

- All 32.4M papers
- Dense ANN via HNSW, joined with graph metrics and communities
- One SQL query composes retrieval + filters + PageRank
- Good at: topical search, hybrid RRF, structural filters

Qdrant — the recommendation move

- Pilot subset (top-N by PageRank) with rich payload
- Multi-example recommendation: 'more like these, less like those'
- Payload filters: year, doctype, community, arxiv class
- Good at: the one move pgvector can't cleanly do in SQL

One MCP tool surface (find_similar_by_examples). Two backends. The agent picks the move by name — not by where the vectors live.

Tool design for agent navigation of scientific literature

Principles from watching agents actually use these tools.

Fewer, composable tools > many specific tools

Agent tool-selection accuracy degrades past a point. Prefer a small set of primitives the agent can combine over a catalog of bespoke endpoints.

Stable, typed schemas

JSON, no free-text fields that need parsing. Every response composes cleanly with whatever the agent does next.

Cursor-based pagination

Never return 10,000 bibcodes in one payload. Give the agent a cursor and let it ask for more if it needs more.

Return compact stubs, full text on request

bibcode · title · first_author · year · citation_count · snippet. Full paper only when the agent asks for it. Context is the scarcest resource.

Include provenance

Which tool returned this? Which query? The agent can reason about its own trail — and so can you, when you read the trace.

Idempotent reads, explicit side effects

Same inputs → same outputs. Caching is trivial. Anything that writes state is called out in the tool name.

Unlocks: exploration modes you can't do with search alone

Each of these is a move an agent takes — not a UI the human drives.

Bridge papers between communities

'Which papers are cited by both the exoplanet community and the atmospheric-chemistry community?' — a single structural query, not a month of hallway conversations.

Citation chains (what this claim depends on)

Follow the chain of results the current paper rests on. Where does the chain terminate — a dataset? a theorem? an assumption someone flagged?

Co-citation and bibliographic coupling

Find papers that are read-alike (cited together) or reference-alike (share references). These are invisible-college detectors.

Temporal evolution

Which subfields are growing? Which are consolidating into review papers? Which had a flurry five years ago and went quiet?

Multi-lens community assignment

Three lenses — citation, semantic, taxonomic — and their disagreements are informative. A paper that's semantically central but citationally isolated is often an 'outsider with a good idea.'

Three community lenses — and their informative disagreements

Each lens has clean semantics. The agent picks the one that fits the question.

Citation

behavioral affinity

- Leiden on the full citation graph
- Three resolutions: ~20 · ~200 · ~2000
- Coverage: 62% of papers
- 'Papers that cite each other'
- Finds invisible-college patterns

Semantic

topical affinity

- k-means on INDUS embeddings
- Three resolutions
- Coverage: 99% (all embedded)
- 'Papers with similar content'
- Catches cross-citing subfields

Taxonomic

institutional affinity

- arXiv class + UAT concepts
- Hierarchical, human-curated
- Coverage: ~8% (arXiv subset)
- 'Papers in the same field'
- Sanity-check for the other two

A paper that's semantically central but citationally isolated is usually an outsider with a good idea — or a review the field hasn't caught up to.

Live demo

A research copilot maps the exoplanet-atmospheres landscape.

User prompt

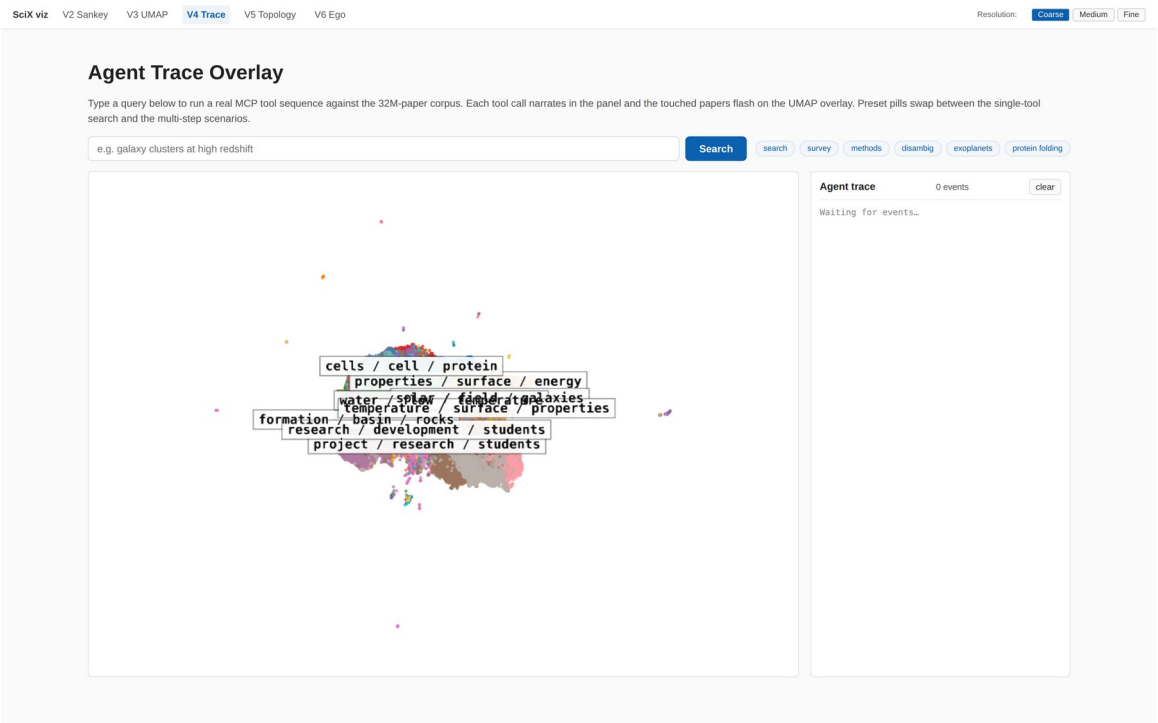
'I'm preparing a grant on exoplanet atmospheric characterization. Map the research landscape — major subcommunities, foundational papers, recent bridges between communities, and what's worth reading first.'

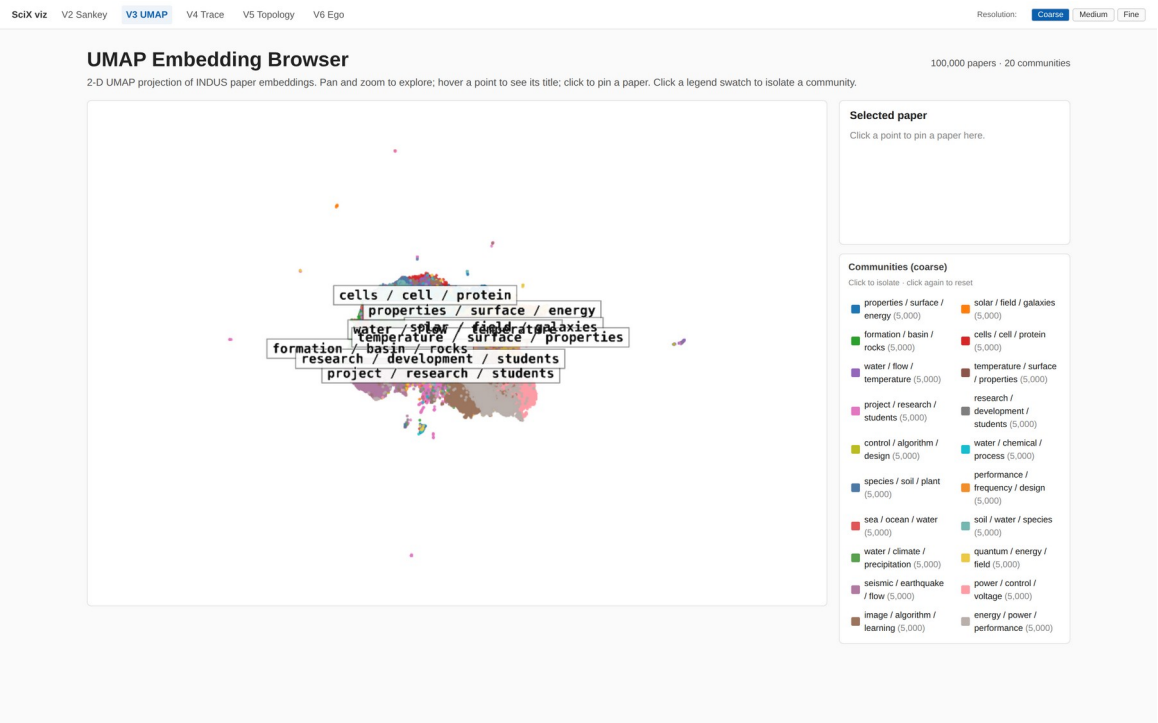
Moves the copilot makes

1	**search**	broad hybrid hit on 'exoplanet atmospheric characterization'
2	**graph_context**	pull community assignments + PageRank for the top hits
3	**citation_similarity**	expand with co-citation bridges
4	**temporal_evolution**	see which communities are growing
5	**find_gaps**	bridge papers between working-set communities
6	**read_paper**	deep-dive on 3 foundational candidates

What the copilot sees while it works

Agent-trace overlay · UMAP of 32M INDUS embeddings.





These are the structural views an agent reasons over — not visually, but as graph edges, community IDs, and embedding coordinates.

What this means if you're using a research copilot

Most of you will be integrating one, not building the knowledge layer. The questions to ask are the same.

What does the corpus cover?

Full field or a rolling window? Do the graph metrics it returns mean what you think they mean?

Does it return structured objects?

A bibcode, not a snippet. IDs compose; prose doesn't. Tools that return prose force the agent to parse its own context.

What moves does it support beyond search?

Citation chains, co-citation, community bridges, temporal slices. If the only primitive is search, you have a faster librarian — not a navigation layer.

Can you trace what it consulted?

Every answer should come with the tool calls it made. This is how you trust it, debug it, and teach it.

How does it complement, not replace, your judgement?

Use it to expand the landscape and surface blind spots. Reserve peer review, interpretation, and prioritization for the human.

Takeaway

From search → **to understanding**

From retrieval → **to navigable scientific context**

From serendipity → **to systems**

Scientific knowledge gets more useful when we treat it like the system it already is — data, methods, results, code, claims — and let agents navigate the connections we've been leaving to chance.

Thank you

Questions?

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Resources

- Repository — github.com/sjarmak/scix-agent
- Live MCP endpoint — mcp.sjarmak.ai/mcp/ (local-only during workshop)
- Paper (ADASS 2026, preprint) — arXiv (pending)
- Contact — stephanie.jarmak1@gmail.com